Metis: one score for each candidate

Shared backbone and head; candidates may be processed in batches.

Qwen3 Base + ScoreHead

1. Candidate inputs

Shared: q, u

	Text	Suffix
c_1	Rules	R
c_2	Weather	R
c_3	Fees	R

q: refund information

u: context; R: last suffix token

2. Encode and score

Qwen3 Base

Shared weights

LoRA $\Delta\theta$

Readout vectors: h_1, h_2, h_3

$h_i \in \mathbb{R}^D$

Shared MLP head

GELU

W_1, b_1 $\quad$ W_2, b_2

$s_i = \mathrm{MLP}_\phi(h_i)$

3. Task decisions

	Raw scores	Task rule
c_1	2.1	Rank c_1, c_3, c_2
c_2	−0.8	Choose c_1
c_3	1.4	Multi $\{c_1, c_3\}$

Illustrative scores

Multi-label example: $s_i \geq 0$

Training: loss and parameter update

Scores	Labels (ranking)
2.1	1
−0.8	0
1.4	1

Task loss

$\mathcal{L}(\boldsymbol{s}, \boldsymbol{y})$

Backpropagation

Update LoRA + head

$\Delta\theta\,, \quad \phi = (W_1, b_1, W_2, b_2)$

LoRA example: base weights θ_0 stay fixed; labels enter only the loss. The toolkit also supports full fine-tuning.

Input tokens · Hidden vectors · Head weights · Scores / loss · Labels

IDs follow the scores; no text generation. Colored cells illustrate object types, not measured values.